

**Data Analytics (DA)**

**Module-End -1**

**Module End Assignment:**

**“Data Cleaning-Analysis-Visualization with Excel”**

## Assignment Task Overview

Create and analyse an E-commerce dataset in Excel. The assignment should demonstrate the following:

Dataset Link: [E-commerce store dataset](#)

- Data entry, import, and basic organization using tables
- Data cleaning: find/remove duplicates, handle missing/inconsistent data, correct formatting errors, and validate data
- Remove the column that contains irrelevant data if it exists
- Application of Excel formulas for calculations, summaries, and logical/text operations
- Use the XLOOKUP/VLOOKUP functions to combine the tables.
- Use of Pivot Tables for advanced summarization
- Descriptive statistics (mean, median, mode, standard deviation, etc.) using formulas or Analysis ToolPak
- Data visualization: create at least three chart types, build an interactive dashboard, and use basic PivotCharts

Deliverables:

- Final Excel file with all steps visible (use multiple sheets or tabs)
- Screenshots illustrating each cleaning/processing step (optional but encouraged)
- One-page summary explaining key findings and decisions in word file

## Skills to be Evaluated and Mapping

| Assignment Section             | Core Skill       | Sub-Skill                             | Expected Proficiency |
|--------------------------------|------------------|---------------------------------------|----------------------|
| Data Entry & Organization      | Spreadsheet Mgmt | Table Creation, Import, Entry         | Beginner             |
| Data Cleaning & Transformation | Data Cleaning    | Duplicates, Missing/Inconsistent Data | Intermediate         |

|                                  |                    |                                                           |              |
|----------------------------------|--------------------|-----------------------------------------------------------|--------------|
| Excel Formulas & Functions       | Statistical/Logic  | SUM, IF, COUNTIF, TEXT, Average,xlookup/vlookup functions | Intermediate |
| Descriptive Statistics           | Data Analysis      | Mean, SD, Mode, Analysis ToolPak                          | Intermediate |
| Data Summarization (Pivot Table) | Data Summarization | Creating, Connecting Tables                               | Intermediate |
| Data Visualization               | Visualization      | Bar, Column, Pie, Sparklines, Dashboards                  | Intermediate |
| Reporting & Insights             | Communication      | Clear Findings Summary                                    | Beginner     |

## Rubric & Mark Allocation (Total 25 Marks)

| Skill/Component                | Full Marks | Criteria for Full Marks                                                                                              | Partial Credit Description                                   | Partial Credit Marks Range |
|--------------------------------|------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------|
| Data Entry & Table Setup       | 3          | Clean tables; correct import; data types properly formatted                                                          | Basic setup, but some improper import/types or minor errors  | 1 – 2                      |
| Data Cleaning & Transformation | 5          | All cleaning steps correct; missing/inconsistent fixed, combining the tables if necessary; and it is well-documented | Most steps correct; minor errors or incomplete documentation | 2 – 4                      |

|                            |   |                                                                    |                                                         |       |
|----------------------------|---|--------------------------------------------------------------------|---------------------------------------------------------|-------|
| Excel Formulas & Functions | 4 | Correct use of SUM, IF, COUNTIF, TEXT; comprehensive application   | Some formulas applied, but inconsistent/partial         | 2 – 3 |
| Descriptive Statistics     | 4 | Accurate calculations; used Analysis ToolPak/similar tool properly | Most stats correct; process partially shown             | 2 – 3 |
| Data Summarization (Pivot) | 3 | Created/used Pivot Table to summarize multi-dimensional data       | Some fields summarized; minor or moderate issues        | 1 – 2 |
| Data Visualization         | 4 | ≥3 chart types; clear labels; dashboards interactive/organized     | Charts missing/labels unclear/dashboards basic          | 1 – 2 |
| Reporting & Insights       | 2 | Clear, concise one-page summary linking findings to visuals        | Partial summary, omits some findings or interpretations | 1 – 1 |

## DA Skill Taxonomy Structure

- Spreadsheet Management
  - Data Entry, Import, Table Creation
- Data Cleaning
  - Duplicates Removal, Missing Data Handling, Consistency Checks
- Formulas & Analysis
  - SUM, IF, COUNTIF, AVERAGE, TEXT Functions, Statistical Measures
- Data Summarization
  - Pivot Tables, Filtering, Sorting
- Visualization

- Charts, Dashboards, PivotCharts
- Communication
  - Summary, Reporting Key Insights

## **Evaluation Guidelines**

1. Open and review all Excel sheets/tabs provided.
2. For each section, match student work to the specific skill and sub-skill area in the taxonomy.
3. Assign full or partial marks per the rubric, using suggested partial mark ranges for “partially complete” work.
4. For ambiguous or incomplete submissions, note what’s missing and correspondingly reduce credit.
5. Comments must explain strengths and improvement points for each section.
6. Objective assessment: focus on function, correctness, clarity, not on visual design beyond the stated requirements.

## **Achievement Criteria & Passing Rules**

- At least partial (not zero) credit must be received in every section—no component can be left unattempted.
- Students must fully achieve at least two major skills: Data Cleaning & Transformation and Data Visualization.
- Feedback and a detailed scorecard must be given for all evaluated assignments.